

C921

Assessment and Evaluation Strategies for Measuring Student Learning

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A nurse educator at a school with a new testing center gives the same dosage-calculation exam twice, a week apart, and students score nearly the same both times. A review then shows the exam mostly measures reading speed, not calculation skill. The testing center has 60 computers. Which statement is accurate?

- A. It cannot be used because the center has only 60 computers.
- B. It is reliable but not valid for measuring calculation skill.
- C. It is valid because scores are consistent.
- D. It is valid but not reliable.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which statement about reliability and validity is true?

- A. Reliability and validity mean the same thing.
- B. A valid test can be unreliable.
- C. A reliable test is always valid.
- D. A test must be reliable to be valid, but reliability does not guarantee validity.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A 50-item exam, given in a large classroom with new desks, was taken by 40 students. Item 12 was answered correctly by 30 students. The class average was 82%. What is the difficulty index (p) for item 12?

- A. 0.60
- B. 0.82
- C. 0.75
- D. 0.25

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

An item had a difficulty index of 0.35 on a test taken by 60 students. How many students answered it correctly?

- A.** 35
- B.** 21
- C.** It cannot be found without upper and lower group data.
- D.** 39

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

An educator whose program recently switched learning-management vendors analyzes one item. In the upper-scoring group of 20 students, 8 answered correctly; in the lower group of 20, 14 answered correctly. The item was about heart-failure medications. What does the discrimination index show?

- A.** $D = -6$; review the item.
- B.** 0.55; a moderately difficult item.
- C.** $D = -0.30$; the item discriminates negatively and should be reviewed.
- D.** $D = +0.30$; a good item.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Midway through a cardiac unit taught in a classroom with a new smart board, an educator gives an ungraded 5-question quiz to find which concepts students are missing so she can adjust tomorrow's lesson. The quiz takes 10 minutes. What type of assessment is this?

- A.** Formative.
- B.** Summative.
- C.** Norm-referenced.
- D.** Admission placement testing.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A comprehensive final exam determines whether students pass the course. What type of assessment is it?

- A.** Diagnostic.
- B.** Summative.
- C.** Formative.
- D.** Formative, because students get their scores back.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A nurse educator at a program with a growing waitlist builds a 100-item medical-surgical final. About 30% of course time and outcomes covered respiratory content, but her draft has 55 respiratory items because those were easiest to write. A colleague reviewed the draft over coffee. What is the main problem?

- A.** Weak content validity; a blueprint should weight about 30 items to respiratory content.
- B.** The reliability coefficient must be too low.
- C.** The exam needs 30% more items overall.
- D.** No problem, since more items always raise reliability.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A faculty team at a school with a new simulation wing reviews a 15-item unit quiz with a KR-20 of 0.52. They want scores reliable enough to support decisions. The quiz was given on a Monday morning. Which change most likely raises reliability?

- A.** Cut items to make it shorter.
- B.** Give it on a different day.
- C.** Nothing; 0.52 is already excellent.
- D.** Add more well-written items on the same content.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A nursing program that recently won a state grant requires students to score at least 90% on a dosage-calculation test to pass, no matter how classmates score. The test is given in the computer lab. What kind of score interpretation is this?

- A.** Norm-referenced.
- B.** Curved grading.
- C.** Criterion-referenced.
- D.** Ipsative assessment.

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
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11. [TRANSFER — reversed stem / different variable]

An instructor gives an A to the top 10% of scorers regardless of their raw scores. What kind of interpretation is this?

- A.** Norm-referenced.
- B.** Every student can earn an A under this rule.
- C.** Standards-based mastery grading.
- D.** Criterion-referenced.

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
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12.

Two clinical instructors at a hospital-based program with a new parking garage grade the same student's care plan and give very different scores. Both have over 10 years of experience. What is the best fix?

- A.** Switch to pass/fail with no criteria.
- B.** Average the two scores from now on.
- C.** Let the more senior instructor grade alone.
- D.** Use a clear analytic rubric with specific criteria and calibrate the raters together.

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A nurse educator at a school ~~with a new testing center~~ gives the same dosage-calculation exam twice, a week apart, and students score nearly the same both times. A review then shows the exam mostly measures reading speed, not calculation skill. ~~The testing center has 60 computers.~~ Which statement is accurate?

Target word: judge reliability and validity **Decides:** Consistent scores, but the wrong skill is measured.

A	Camouflage	Seat count is about practicality, not the validity problem.
B	Correct	Consistency shows reliability; measuring reading speed undermines validity.
C	Confusion set	Consistency is reliability, not validity.
D	Inverted operation	Reverses both findings.

2. Answer D Concept 1 • transfer

Which statement about reliability and validity is true?

Target word: relationship **Decides:** Reliability is necessary but not sufficient.

A	Confusion set	They are distinct qualities.
B	Inverted operation	Validity requires reliability.
C	Absolute language	A test can be consistent and still measure the wrong thing.
D	Correct	Inconsistent scores can't measure accurately, but consistent ones can still miss the target.

3. Answer C Concept 2

A 50-item exam, ~~given in a large classroom with new desks,~~ was taken by 40 students. Item 12 was answered correctly by 30 students. ~~The class average was 82%.~~ What is the difficulty index (p) for item 12?

Target word: difficulty index **Decides:** 30 correct / 40 takers.

A	Confusion set	Divides by the 50 test items instead of the 40 students.
B	Camouflage	The class average describes the whole test, not item 12.
C	Correct	$30 / 40 = 0.75$.
D	Inverted operation	That is the proportion who missed it.

4. Answer B Concept 2 • transfer

An item had a difficulty index of 0.35 on a test taken by 60 students. How many students answered it correctly?

Target word: solve for number correct **Decides:** 0.35×60 .

A	Wrong stem number	Treats the index as a count.
B	Correct	$0.35 \times 60 = 21$.
C	Adjacent category	Group data are needed for discrimination, not difficulty.
D	Inverted operation	That is the number who missed it.

5. Answer C Concept 3

An educator ~~whose program recently switched learning management vendors~~ analyzes one item. In the upper-scoring group of 20 students, 8 answered correctly; in the lower group of 20, 14 answered correctly. ~~The item was about heart failure medications.~~ What does the discrimination index show?

Target word: compute and interpret D **Decides:** $8/20 - 14/20 = 0.40 - 0.70$.

A	Dropped element	Uses raw counts instead of proportions.
B	Adjacent category	That is the difficulty index $((8 + 14) / 40)$, not discrimination.
C	Correct	Low scorers did better than high scorers, a sign of a flawed item or key.
D	Inverted operation	Subtracts in the wrong order.

6. Answer A Concept 4

Midway through a cardiac unit ~~taught in a classroom with a new smart board~~, an educator gives an ungraded 5-question quiz to find which concepts students are missing so she can adjust tomorrow's lesson. ~~The quiz takes 10 minutes.~~ What type of assessment is this?

Target word: classify assessment **Decides:** Used to adjust teaching during learning.

A	Correct	It guides instruction while learning is in progress.
B	Inverted operation	Summative judges achievement at the end.
C	Confusion set	That describes score interpretation, not purpose.
D	Adjacent category	Placement happens before a program, not mid-unit.

7. Answer B Concept 4 • transfer

A comprehensive final exam determines whether students pass the course. What type of assessment is it?

Target word: classify — reversed **Decides:** End-point decision.

A	Adjacent category	Diagnostic assessment identifies needs before instruction.
B	Correct	It judges achievement at the end for a decision.
C	Inverted operation	Formative guides learning along the way.
D	Confusion set	Returning scores does not change its purpose.

8. Answer A Concept 5

A nurse educator at a program ~~with a growing waitlist~~ builds a 100-item medical-surgical final. About 30% of course time and outcomes covered respiratory content, but her draft has 55 respiratory items because those were easiest to write. ~~A colleague reviewed the draft over coffee.~~ What is the main problem?

Target word: name the flaw **Decides:** Item weighting doesn't match content emphasis.

A	Correct	Items should mirror the emphasis of what was taught.
B	Confusion set	The problem is what is sampled, not score consistency.
C	Wrong stem number	Misreads the 30% emphasis as a length change.
D	Adjacent category	Reliability does not fix a skewed content sample.

9. Answer D Concept 6

A faculty team at a school ~~with a new simulation wing~~ reviews a 15-item unit quiz with a KR-20 of 0.52. They want scores reliable enough to support decisions. ~~The quiz was given on a Monday morning.~~ Which change most likely raises reliability?

Target word: raise reliability **Decides:** Short test, low coefficient.

A	Inverted operation	Shorter tests usually lower reliability.
B	Camouflage	The day does not explain low internal consistency.
C	Confusion set	About 0.70 or higher is the usual target.
D	Correct	Longer tests of homogeneous content are more reliable.

10. Answer C Concept 7

A nursing program ~~that recently won a state grant~~ requires students to score at least 90% on a dosage-calculation test to pass, no matter how classmates score. ~~The test is given in the computer lab.~~ What kind of score interpretation is this?

Target word: classify interpretation **Decides:** Fixed standard, independent of peers.

A	Inverted operation	Norm-referenced compares students with each other.
B	Confusion set	Curving adjusts to the group, which this policy rules out.
C	Correct	Scores are compared with a fixed standard.
D	Adjacent category	Ipsative compares a student with his or her own past performance.

11. Answer A Concept 7 • transfer

An instructor gives an A to the top 10% of scorers regardless of their raw scores. What kind of interpretation is this?

Target word: classify — reversed **Decides:** Compared with other students.

A	Correct	Grades depend on rank within the group.
B	Absolute language	Only the top 10% can.
C	Confusion set	Mastery grading uses fixed criteria.
D	Inverted operation	No fixed standard is used.

12. Answer D Concept 8

Two clinical instructors at a hospital-based program ~~with a new parking garage~~ grade the same student's care plan and give very different scores. ~~Both have over 10 years of experience.~~ What is the best fix?

Target word: improve scoring consistency **Decides:** Scores differ between raters.

A	Adjacent category	Removing criteria makes scoring less consistent.
B	Dropped element	Hides the disagreement without fixing its cause.
C	Camouflage	Experience does not ensure consistent scoring.
D	Correct	Shared criteria and practice scoring raise inter-rater reliability.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	2	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Validity vs. reliability	Reliability is consistency; validity is measuring what you intend. A test must be reliable to be valid, but a reliable test can still be invalid.	Treating consistent scores as proof of validity.	Consistent scores on a test that measures the wrong thing.
2	Item difficulty (p)	p = number who answered correctly / number who took the item; higher p means an easier item.	Using the number of test items or the class average.	A count correct and a count of test-takers.
3	Item discrimination (D)	D = proportion correct in the upper group – proportion correct in the lower group; a negative D flags a flawed item.	Subtracting in the wrong order or using raw counts.	Upper and lower group results for one item.
4	Formative vs. summative assessment	Formative assessment guides learning while it is under way; summative assessment judges achievement at the end.	Calling any scored quiz summative.	The purpose: adjust teaching vs. make a final decision.
5	Test blueprint and content validity	A blueprint weights items to match the content and outcomes taught, supporting content validity.	Writing the most items where they are easiest to write.	Item counts that don't match teaching emphasis.
6	Raising reliability	Adding well-written items on the same content usually raises reliability (KR-20 or α); about 0.70 or higher is a common target for classroom tests.	Shortening the test or changing test day to raise reliability.	A low reliability coefficient on a short test.
7	Norm- vs. criterion-referenced interpretation	Criterion-referenced scores are compared with a fixed standard; norm-referenced scores are compared with other test-takers.	Mixing up the two, or assuming any cutoff is norm-referenced.	"No matter how classmates score" vs. "top 10%."
8	Inter-rater reliability	Clear analytic rubrics and rater calibration make scores consistent across graders.	Letting the most senior grader decide alone or averaging forever.	Two graders giving very different scores for the same work.

Course scope (public description): Assessment and Evaluation Strategies for Measuring Student Learning addresses the academic nurse educator's role in the design, development, implementation, and evaluation of student achievement outcomes in nursing education programs. This course requires students to integrate best practices from nursing theory and theories of learning to assess student learning in diverse educational settings. Topics include validity, reliability, and practicality of assessments, interpreting item difficulty and discrimination test results, and analyzing student achievement and learning outcomes data. This course has no prerequisites.